`GeoModels` Tutorial: simulation, estimation and prediction of positive spatial data using Weibull random fields

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Introduction

In this tutorial we show how to analyze geo-referenced spatial data with positive support using Weibull random fields (RFs) [Bevilacqua et al., 2020] with the R package `GeoModels` [Bevilacqua et al., 2026]. The Weibull distribution is a flexible parametric model for positive data allowing both right and left skewness.

We first load the R libraries needed for the analysis and set the name of the model in the `GeoModels` package:

```
rm(list=ls())
require(GeoModels)
require(fields)
require(hypergeo)
set.seed(24)
model="Weibull" # model name in the GeoModels package
```

Simulation of Weibull random fields

The definition of a Weibull RF starts by considering a 'parent' Gaussian RF $Z = \{Z(\boldsymbol{s}), \boldsymbol{s} \in S\}$, where $\boldsymbol{s}$ represents a location in the domain S. In this tutorial, we assume $S = [0,1]^2 \subseteq \mathbb{R}^2$ and that Z is stationary with zero mean, unit variance and correlation function $\rho(\boldsymbol{h}) = \text{cor}\{Z(\boldsymbol{s}+\boldsymbol{h}), Z(\boldsymbol{s})\}$.

Given Z_1, Z_2, two independent copies of Z, a RF $U = \{U(\boldsymbol{s}), \boldsymbol{s} \in S\}$ with marginal distribution $Weibull(\kappa, \nu(\kappa))$ can be derived by the transformation

$$U(\boldsymbol{s}) = \nu(\kappa)\left(\frac{1}{2}\sum_{k=1}^{2} Z_k(\boldsymbol{s})^2\right)^{1/\kappa}, \tag{1}$$

where $\nu(\kappa) = \Gamma^{-1}(1+1/\kappa)$, $\kappa > 0$ is a shape parameter and $\Gamma(\cdot)$ is the gamma function. Under this specific parametrization, $\text{E}\{U(\boldsymbol{s})\} = 1$, $\text{var}\{U(\boldsymbol{s})\} = \Gamma(1+2/\kappa)\nu^2(\kappa) - 1$ and the correlation function is given by:

$$\rho_U(\boldsymbol{h}) = \frac{\nu^{-2}(\kappa)}{\Gamma(1+2/\kappa) - \nu^{-2}(\kappa)}\left[{}_2F_1\left(-\frac{1}{\kappa}, -\frac{1}{\kappa}; 1; \rho^2(\boldsymbol{h})\right) - 1\right]. \tag{2}$$

Here ${}_2F_1(a,b;c;x)$ is the Gaussian hypergeometric function [Abramowitz and Stegun, 1970]. In the `GeoModels` package it is computed using the function `hypergeo` of the `hypergeo` package [Hankin, 2016].

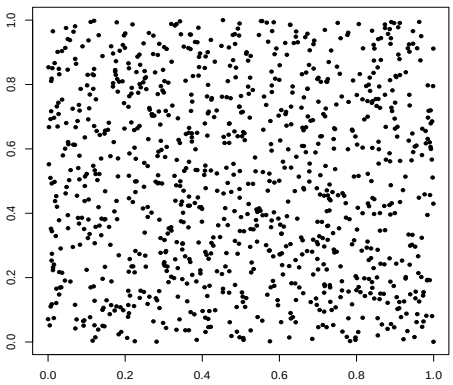
Then a non-stationary version can be defined as:

$$W(\boldsymbol{s}) = \mu(\boldsymbol{s})U(\boldsymbol{s}), \qquad \mu(\boldsymbol{s}) > 0. \tag{3}$$

In this case $\mathrm{E}\{W(\boldsymbol{s})\} = \mu(\boldsymbol{s})$, $\mathrm{var}\{W(\boldsymbol{s})\} = \mu(\boldsymbol{s})^2\{\Gamma(1+2/\kappa)\nu^2(\kappa) - 1\}$, and a spatial regression model can be obtained by assuming that $\mu(\boldsymbol{s}) = \exp\{X(\boldsymbol{s})^T\boldsymbol{\beta}\}$ where $X(\boldsymbol{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_{k-1})^T$ is a k-dimensional vector of (unknown) parameters.

Thus, in order to obtain a realization from a Weibull RF we need to specify regression mean parameters, a shape parameter and a parametric correlation model for $\rho(\boldsymbol{h})$. We first set the spatial coordinates:

```
N=1000 # number of location sites
x = runif(N,0,1)
y = runif(N,0,1)
coords=cbind(x,y) # spatial coordinates
plot(coords,pch=20,xlab="",ylab="")
```



Then we fix $k = 2$ and we build the matrix of covariates and fix the regression mean parameters:

```
X=cbind(rep(1,N),runif(N)) # matrix covariates
mean=-0.3; mean1=0.5 # regression parameters
```

where `mean` and `mean1` are respectively β_0 and β_1.

The names of the marginal parameters associated with the Weibull model can be obtained with the function `NuisParam` (note that the option `num_betas` is the number of regression parameters involved):

```
NuisParam(model,num_betas=2)
# [1] "mean" "mean1" "nugget" "shape"
```

For the correlation function we assume a special case of the isotropic Generalized Wendland class [Bevilacqua et al., 2019], i.e. the Askey model:

$$\rho(\boldsymbol{h};\alpha,\delta) = \begin{cases} (1-\|\boldsymbol{h}\|/\alpha)^{\delta}, & \|\boldsymbol{h}\| < \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Using asymptotic arguments Bevilacqua et al. [2019] show that this correlation model has the same features of the exponential correlation model. Additionally, it is compactly supported, an interesting feature from the computational point of view. We set the Askey model and the associated parameters. Note that the function `CorrParam` returns parameter names associated with a given correlation model:

```
corrmodel="Wend0" ## correlation model
CorrParam(corrmodel) ## names of correlation model parameters
# [1] "power2" "scale"
scale=0.2
power2=4
```

Here the `scale` parameter corresponds to α, the compact support of the correlation model. Finally, we set the shape parameter of the Weibull RF and the nugget parameter:

```
shape=2 # shape of the Weibull RF
nugget=0 # nugget parameter
```

We are now ready to simulate a Weibull random field using the function `GeoSim`:

```
param=list(mean=mean,mean1=mean1,nugget=nugget,
          scale=scale,power2=power2,shape=shape)
set.seed(312)
data=GeoSim(coordx=coords,corrmodel=corrmodel,
                    model=model,param=param,X=X)$data
```

The simulation is performed using Cholesky decomposition for the two Gaussian RFs involved (see Equation (1)).

Estimation of Weibull random fields

The density of the bivariate random vector $(U(\boldsymbol{s}_i), U(\boldsymbol{s}_j))^T$ is given by Bevilacqua et al. [2020]:

$$f_U(u_i, u_j) = \frac{\kappa^2 (u_i u_j)^{\kappa-1}}{\nu(\kappa)^{2\kappa}(1-\rho_{ij}^2)} \exp\left\{-\frac{u_i^\kappa + u_j^\kappa}{\nu(\kappa)^\kappa (1-\rho_{ij}^2)}\right\} \times I_0\left(\frac{2|\rho_{ij}|(u_i u_j)^{\kappa/2}}{\nu(\kappa)^\kappa(1-\rho_{ij}^2)}\right), \tag{4}$$

where $I_a(x)$ denotes the modified Bessel function of the first kind of order a, and $\rho_{ij} = \rho(\boldsymbol{s}_i - \boldsymbol{s}_j)$. The bivariate densities of W can be derived from (4) as:

$$f_W(w_i, w_j) = (\mu_i \mu_j)^{-1} f_U\left(\frac{w_i}{\mu_i}, \frac{w_j}{\mu_j}\right). \tag{5}$$

Given the observations $w(\boldsymbol{s}_1), \ldots, w(\boldsymbol{s}_N)$, the pairwise log-likelihood is defined as:

$$pl(\boldsymbol{\theta}) = \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \log\{f_W(w_i, w_j)\}\, c_{ij},$$

where $\boldsymbol{\theta} = (\beta_0, \beta_1, \kappa, \alpha, \delta)^T$ and $w_i = w(\boldsymbol{s}_i)$ for notation convenience.

An efficient way to specify the (non-symmetric) weights from a computational viewpoint is based on neighborhoods:

$$c_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Here $N_k(\boldsymbol{s}_l)$ is the set of the k nearest neighbors of the point $\boldsymbol{s}_l$. The pairwise likelihood estimator $\hat{\boldsymbol{\theta}}_{pl}$ is obtained by maximizing $pl(\boldsymbol{\theta})$ with respect to the free parameters of $\boldsymbol{\theta}$. In the `GeoModels` package we can choose the fixed parameters and the parameters that must be estimated. Pairwise likelihood estimation is performed with the function `GeoFit`:

```
start=list(mean=mean,mean1=mean1,scale=scale,shape=shape)
fixed=list(nugget=nugget,power2=power2)
# Maximum composite-likelihood fitting of the Weibull random field:
fit=GeoFit(data=data,coordx=coords,corrmodel=corrmodel,
    model=model,X=X,optimizer="BFGS",start=start,fixed=fixed,
    neighb=3)
```

The object `fit` includes information about the pairwise likelihood estimation:

```
fit
# Historical output from the original tutorial (2024):
# Setting: Marginal Composite-Likelihood
# Model: Weibull; covariance model: Wend0; optimizer: BFGS
# Number of spatial coordinates: 1000
# Maximum log-Composite-Likelihood value: -3693.89
# Estimated parameters:
# mean mean1 scale shape
# -0.2418 0.3988 0.2267 1.9590
```

Note that the option `neighb=3` sets the neighborhood size of the weight function (6), i.e. $k = 3$; it does not set the compact support of the spatial correlation model.

Checking model assumptions

Given the estimation of the mean $\widehat{\mu(\boldsymbol{s})} = \exp\{X_1(\boldsymbol{s})\hat{\beta}_0 + X_2(\boldsymbol{s})\hat{\beta}_1\}$, the estimated residuals

$$\widehat{u(\boldsymbol{s}_i)} = \frac{w(\boldsymbol{s}_i)}{\widehat{\mu(\boldsymbol{s}_i)}}, \qquad i = 1, \ldots, N, \tag{7}$$

can be viewed as a realization of U, a stationary RF with Weibull marginal distribution, unit mean and correlation function given by (2). The estimated residuals can be computed using `GeoResiduals`:

```
res=GeoResiduals(fit) # computing residuals
```

Then the agreement of the marginal distribution assumption on the residuals with the theoretical model can be graphically checked with the `GeoQQ` function (Figure 1, left part):

```
GeoQQ(res)
GeoQQ(res,type="D")
```

The covariance model assumption can be checked comparing the empirical and the estimated semivariogram using the `GeoVariogram` and `GeoCovariogram` functions (Figure 1, right part):

```
vario=GeoVariogram(data=res$data,coordx=coords,
                           maxdist=0.3) # empirical variogram
GeoCovariogram(res,show.vario=TRUE,vario=vario,pch=20)
```

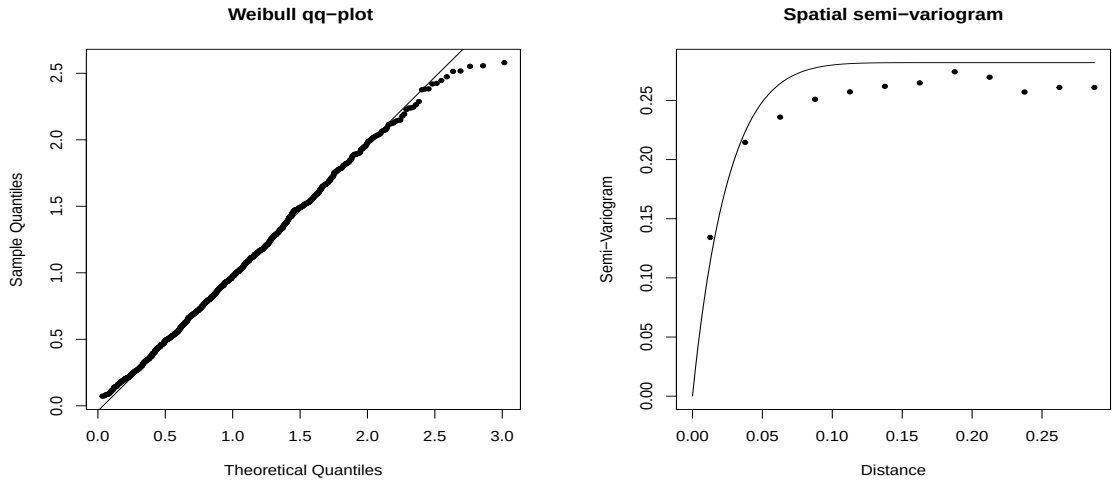


Figure 1: Left: QQ-plot for the Weibull model residuals. Right: empirical vs estimated semi-variogram function for the residuals.

Prediction of Weibull random fields

The optimal linear prediction of a Weibull RF at a location $\boldsymbol{s}_0$ is given by Bevilacqua et al. [2020]:

$$\widehat{W}(\boldsymbol{s}_0) = \widehat{\mu}(\boldsymbol{s}_0)\left(1 + \sum_{i=1}^{N} \widehat{\lambda}_i[\widehat{U}(\boldsymbol{s}_i) - 1]\right), \tag{8}$$

where the vector of weights $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)^T$ is given by $\boldsymbol{\lambda} = \boldsymbol{R}^{-1}\boldsymbol{c}$. Here the cross-correlation vector and the correlation matrix are, respectively,

$$\boldsymbol{c} = \big(\text{cor}\{U(\boldsymbol{s}_0), U(\boldsymbol{s}_1)\}, \ldots, \text{cor}\{U(\boldsymbol{s}_0), U(\boldsymbol{s}_N)\}\big)^T, \quad \boldsymbol{R} = [\text{cor}\{U(\boldsymbol{s}_i), U(\boldsymbol{s}_j)\}]_{i,j=1}^{N}.$$

Both are computed using (2).

We first set the spatial locations to predict and the associated covariates. In this example, we choose a regular fine grid in order to construct a prediction map:

```
xx=seq(0,1,0.013)
loc_to_pred=as.matrix(expand.grid(xx,xx))
Nloc=nrow(loc_to_pred)
Xloc=cbind(rep(1,Nloc),runif(Nloc))
```

Then the optimal linear prediction (8), using the estimated parameters, can be performed using the `GeoKrig` function:

```
pr=GeoKrig(fit,loc=loc_to_pred,Xloc=Xloc,mse=TRUE,
                        sparse=TRUE)
```

Note that the option `sparse=TRUE` allows specific algorithms for sparse matrices implemented in the **spam** package [Gerber et al., 2017, Furrer and Sain, 2010] when performing the linear algebra associated with the covariance matrix.

We can compare the map of simulated data with the kriging prediction (and associated mean square error) with the following code (see Figure 2):

```
par(mfrow=c(1,3))
quilt.plot(coords,data,main="Data")
quilt.plot(loc_to_pred,pr$pred,xlab="",ylab="",
          main="Kriging")
quilt.plot(loc_to_pred,pr$mse,xlab="",ylab="",
          main="MSE")
```

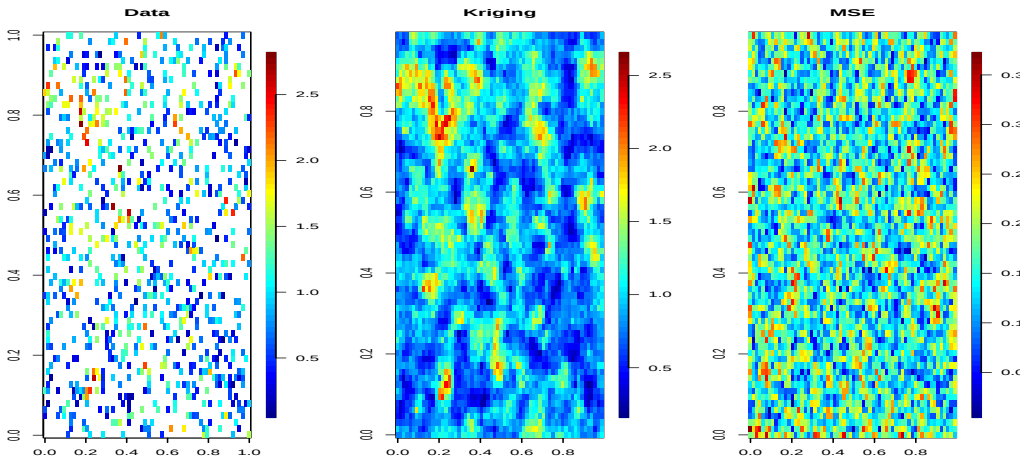


Figure 2: From left to right: colored map of observed data, kriging prediction and associated mean squared error.

References

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